Name: ______________________

Student ID: __________________

Signature: ___________________

UNSW, Sydney

School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

Term 1, 2024

ELEC3115 - Electromagnetic Engineering

1. Time allowed: 2 (two) hours.
2. Reading Time: 10 (ten) minutes.
3. Total number of questions set in this paper: 5 (five). Candidates must answer **ALL five** questions: **THREE** from Part A and **TWO** from Part B.
4. Marks for each question are indicated at the right-hand side of each question. This examination paper carries 100 marks; it contributes 50% of the total course assessment**.**
5. This paper contains 7 pages.
6. Answer each question in a **separate answer book**. Write your name, student ID and Question number on the front page of **each answer book**.
7. This paper **MAY NOT** be retained by the candidate.
8. This is a closed book examination. Students may only bring the following materials into the examination room:
   - UNSW-approved electronic calculators.
   - Drawing instruments.
9. Assumptions made in answering the questions should be stated explicitly.
10. All answers must be written in ink. Except where they are expressly required, pencils may only be used for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

(a)[9 marks]

(i) [4 marks] What is polarization of a dielectric material and how do you relate it to the polarization vector?

(ii)[5 marks] Using the relation of the polarization vector $\vec{P} = \chi_e\left(\varepsilon_0 \vec{E}\right)$ and bound charge density $\rho_p = -\vec{\nabla}.\vec{P}$, show that the relative permittivity of a dielectric material is $\varepsilon_r = 1 + \chi_e$, where χ_e is the electric susceptibility, ε_0 is the permittivity of free space and $\vec{E}$ is the applied electric field.

(b)[6 marks] A high-voltage parallel plate capacitor is made with two aluminium foils of diemensions 35 mm wide and 2.5 m long. An oil-impregnated paper of thickness 0.1 mm insulates the two plates from each other. The relative permittivity of the insulating paper is 2.5, and the dielectric strength is 20 MV/m. Calculate the followings to be included in the datasheet of the capacitor:

(i)[3 marks] capacitance value

(ii)[3 marks] rated voltage (i.e., the maximum voltage that the capacitor can withstand without breakdown).

(c)[5 marks] A customer complained that capacitor given in (b) is faulty because when 15% of the rated voltage was applied, the capacitor's insulating layer showed current conduction (i.e., dielectric breakdown). However, the value of the capacitance nearly matched the datasheet value when measured. What could be the possible cause of such a fault? Why does the value of capacitance match the datasheet value?

QUESTION 2 [20 marks]

(a)[12 marks] In a two-wire transmission line of Figure 1, the two identical cylindrical conductors of length 45 m, radius 0.5 mm run parallel. The centre-to-centre separation between the conductors is 4.5 mm. The surrounding medium is air.

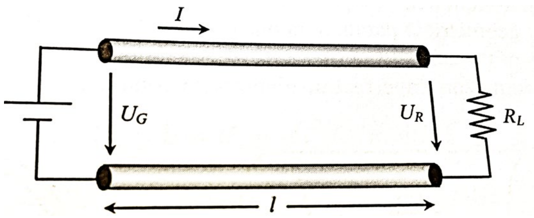

Figure 1

Question 2 continues on the next page

- **(i) [3marks]** Calculate the capacitance per meter between the two conductors. Which method did you use to calculate this capacitance and why?
- **(ii) [3marks]** The line conductors are made of copper with a conductivity of 5.6×10^7S/m. A DC voltage source U_G = 55 V is supplying a load R_L = 55 Ω. Calculate the current I, and load side voltage U_R.
- **(iii) [4 marks]** The current I establishes a magnetic field, and hence inductance also forms in the conductors. The inductance per meter of a cylindrical conductor of radius a can be derived from the internal and external magnetic fields and is given as $L=\left(\frac{\mu_0}{8\pi}+\frac{\mu_0}{2\pi}ln\frac{d}{a}\right)$ H/m. This inductance L is between the centre of the conductor and an arbitrary point located at a radial distance d from the centre of the conductor.

  Calculate the inductance per meter between the given two-wire transmission system which is formed by two identical cylindrical conductors. You may ignore the mutual inductance.
- **(iv) [2 marks]** Draw an electric circuit showing the resistance, capacitance, and inductance (all units in per meter) of the two-wire system of Figure 1.

(b) [8 marks] The presence of capacitance and inductance between the two conductors indicates that both electrostatic and electromagnetic forces act upon the conductors. Considering the centre-to-centre separation distance between the two conductors as a variable x, and assuming x>> a radius of the conductor, calculate

- **(i) [4 marks]** the electrostatic force per meter between the two conductors for the value of x = 4.5 mm, a = 0.5 mm, and the constant voltage V= 55V.
- **(ii) [4 marks]** the electromagnetic force per meter between the two conductors for the value of x = 4.5 mm and the current found in part (a) (iii) above.

Useful differentiations: $\frac{d}{dx}\left(\frac{1}{ln(x/a)}\right)=\frac{1}{x\,ln^2(x/a)};\quad \frac{d}{dx}\left(ln(x/a)\right)=1/x$

QUESTION 3 [20 marks]

(a) [8 marks] Figure 2 shows a magnetic device with two airgaps and two coils. The depth of the upper and lower pieces is the same. The upper piece has identical left and right limbs. The permeability of the magnetic core is infinite.

(i) [3 marks] Draw the magnetic circuit of the device.

Question 3 continues on the next page

(ii)[5 marks] Using the currents, number of turns and the dimensions shown in Figure 2, find an expression for the flux density in the airgaps.

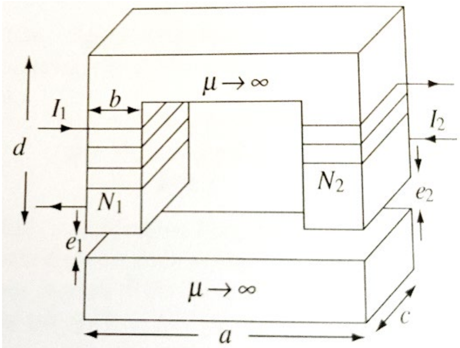

Figure 2

(b)[4 marks] The current I_1 in coil 1 of the Figure 2 is an AC sinusoidal current of frequency f, and induces a sinusoidal magnetic flux density of a peak value B_{max}. Such a time-varying magnetic field induces a voltage in coil 2 by Faraday's law. Show that the root mean square (RMS) value of the voltage induced in the coil 2 is given by

$$V_2 = 4.44 f N_2 B_{\max} S_c$$

(c) [4 marks] The magnetic device of Figure 2 is designed to operate with 240 V(rms) and 50 Hz. For operation in a remote mining site, the device needs to connect to a diesel generator which generates sinusoidal AC voltage at 16.67 Hz. The generator output voltage (RMS) is adjustable. The cross-sectional area of the core $S_c = 0.052\text{m}^2$. The peak flux denisity B_{max} in the core must be limited to 1.1T to prevent magnetic saturation. What should be the voltage of the generator to run the device for the supply frequency 16.67 Hz?

(d) [4 marks] Explain how *eddy currents* occur in the core of the magnetic device shown in Figure 2. What will be the effect of eddy current in the device? How can we reduce this effect?

See next page for question 4

QUESTION 4 [26 marks]

Consider a lossless coaxial cable of length $l = 2$ m, filled by a non-magnetic dielectric.

The cable possesses an inductance per unit length $\mathcal{L} = 250$ nH/m and a capacitance per unit length $\mathcal{C} = 100$ pF/m.

The cable is terminated by an open circuit, and fed by an DC voltage generator with internal resistance $R_G = 950\,\Omega$ and voltage $V_G = 1$ V. An ideal switch is interposed between the generator and the cable, as shown in Figure 4 below.

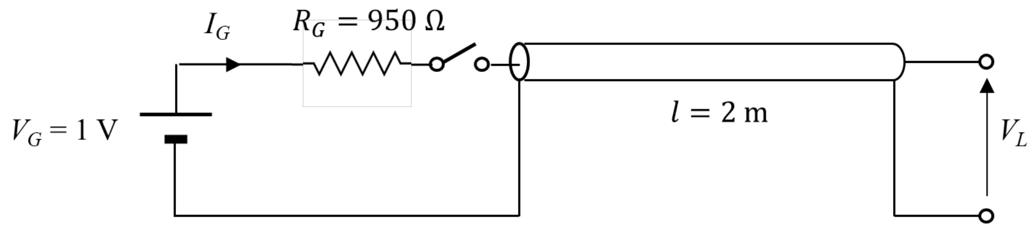

Figure *3*

At time $t = 0$, the switch is closed and the generator is connected to the cable. For the time interval $t = 0 - 60$ ns draw:

(i) [8 marks]The reflection diagram, which must contain the numerical values of the voltage and current on every path, and a correct labelling of the time axis.

(ii) [5 marks]The voltage at the open-circuit termination, $V_L(t)$.

(iii) [5 marks]The current at the generator, $I_G(t)$.

(iv) [4 marks]Construct a lumped-element circuit that best approximates the behaviour of the "real" circuit. Describe the reasoning you use to construct this circuit and provide the numerical value of its parameters.

(v) [4 marks]Draw the voltage $V_L(t)$ in the time interval $t = 0 - 60$ ns as would be obtained from the lumped-elements approximation drawn at point (iv) above. Discuss the difference or similarity between the "true" voltage drawn at point (ii), and the one you obtain here.

See next page for question 5

QUESTION 5 [14 marks]

Consider a rectangular waveguide with transverse dimensions *a* = 7.112 mm and *b* = 3.556 mm. The dielectric inside the waveguide is air ($\varepsilon_r = 1, \mu_r = 1$).

(i) [7 marks] Calculate the cutoff frequency of the lowest five modes of propagation that this waveguide can sustain. Include both TE and TM modes in the count.

(ii) [3 marks] For the mode with the lowest cutoff frequency, calculate the phase velocity and the group velocity for narrow-band signals centred around a frequency $f = 25$ GHz.

(iii) [4 marks] Explain the difference between phase velocity and group velocity. In particular, explain how it is possible that the phase velocity is higher than the speed of light. Use sketches and diagrams to support your argument.

Useful formula and constants for part A:

$$\varepsilon_o = 8.854\times 10^{-12}\ C/m$$

$$\bar{\nabla}\bullet\bar{D} = \frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}$$

$$\bar{\nabla}\times\bar{A} = \bar{a}_x\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)+\bar{a}_y\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)+\bar{a}_z\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)$$

$$\oint_s \bar{E}\bullet d\bar{s} = \frac{Q_{enclosed}}{\varepsilon} \quad or \quad \oint_s \bar{D}\bullet d\bar{s} = Q_{enclosed}$$

$$Q_{enclosed} = \int_v \rho dv \ or \int_s \rho_s ds \ or \int_l \rho_l dl$$

$$V_{ba} = -\int_a^b \bar{E}\bullet d\bar{l}$$

$$\bar{P} = \bar{D}-\varepsilon_0\bar{E},\quad \bar{D}=\varepsilon\bar{E},\quad \varepsilon=\varepsilon_0\varepsilon_r$$

$$\bar{E} = \left|\frac{\rho_l}{2\pi\varepsilon r}\right|\bar{a}_r\ \ V/m,\quad V_{21} = \frac{\rho_l}{2\pi\varepsilon}\ln\left(\frac{r_1}{r_2}\right)$$

$$|E_{1t}| = |E_{2t}|,\quad |D_{1n}| = |D_{2n}|$$

$$C = \frac{\pi\varepsilon}{\ln\frac{d-a}{a}}\ \ F/m,\quad C = \frac{2\pi\varepsilon}{\ln\left(\frac{b}{a}\right)}\ \ F/m$$

$$w_e = \frac{1}{2}\bar{D}\bullet\bar{E}\ [J/m^3],\ \bar{F}_e = \pm\bar{\nabla}W_e$$

$$\bar{\nabla}^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 V}{\partial\varphi^2}+\frac{\partial V^2}{\partial z^2} = 0$$

$$C = \frac{\pi\varepsilon_0}{\ln\left[\left(\frac{d}{2a}\right)+\sqrt{\left(\frac{d}{2a}\right)^2-1}\right]}\ \ F/m$$

$$\mu_0 = 4\pi\times 10^{-7}\ \mathrm{H/m}$$

$$\oint_c \bar{B}\bullet d\bar{l} = \mu_0 I_c,\quad \oint_c \bar{H}\, d\bar{l} = I_{enclosed}$$

$$\bar{B} = \bar{\nabla}\times\bar{A},$$

$$\bar{B} = \mu_o\bar{H}+\mu_o\bar{M},\quad \bar{B} = \mu_0(1+\chi_m)\bar{H} = \mu_0\mu_r\bar{H} = \mu\bar{H}$$

$$\bar{\nabla}^2\bar{A} = -\mu_0\bar{J}$$

$$\Phi = \int_s \bar{B}\bullet d\bar{s},\quad \Re = \frac{l}{\mu S},$$

$$L = \frac{\lambda = N\varphi}{i},$$

$$EMF = -N\frac{d\phi}{dt},\quad EMF = \oint_C\left(\vec{u}\times\vec{B}\right).d\vec{l}$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$V_{rms} = 4.44\, fNB_m S_c$$

$$\delta = \sqrt{\frac{2}{\sigma\mu\omega}},\quad R_{ac} = \frac{1}{\delta\sigma}\left(\frac{l}{\pi(D-\delta)}\right)$$

$$P_h = k_2 B_m^n f,\quad P_e = k_1 B_m^2 f^2$$

$$\vec{J} = \rho\vec{u},\quad I = \int_s \vec{J}\bullet d\vec{s}$$

$$\vec{J} = \sigma\vec{E},\quad R = \frac{l}{\sigma S},\quad P = \int_{v'}\left(\vec{E}\bullet\vec{J}\right)dv'$$

$$\vec{\nabla}\bullet\vec{J} = -\frac{\partial\rho}{\partial t},\quad \rho = \rho_0 e^{-\left(\frac{\sigma}{\varepsilon}\right)t}$$

$$\vec{\nabla}\times\vec{J} = 0$$

$$RC = \frac{\varepsilon}{\sigma}$$

END OF EXAMINATION PAPER